

pypopratio

Adarsh Ranjan

May 2026

1 Introduction

Fine-structure transitions of neutral and singly ionized atoms provide some of the most powerful diagnostics of the physical conditions in diffuse astrophysical gas. Because the relative populations of closely spaced fine-structure levels are sensitive to collisions, radiation fields, and fluorescence processes, they can be used to constrain the density, temperature, thermal pressure, and ambient radiation environment of the interstellar medium (ISM), circumgalactic medium (CGM), and high-redshift absorption systems. Among these species, neutral carbon (C I) is particularly valuable because its ground electronic state contains three fine-structure levels whose populations are readily accessible through ultraviolet absorption spectroscopy (Jenkins & Tripp, 2001).

The relative populations of the C I, C I*, and C I** levels are governed by a balance between spontaneous radiative decay, collisional excitation and de-excitation, direct radiative excitation by the cosmic microwave background (CMB), and indirect excitation through ultraviolet (UV) fluorescence pumping. Since the energy separations between these levels are small, the excitation is especially sensitive to collisions with neutral hydrogen, electrons, and molecular hydrogen at typical ISM temperatures. At low densities, radiative processes dominate, while at sufficiently high densities the level populations approach local thermodynamic equilibrium (LTE). This sensitivity makes fine-structure excitation a robust probe of thermal pressure in diffuse gas (Jenkins & Tripp, 2001).

Early studies demonstrated the diagnostic potential of fine-structure excitation in a wide range of environments. Jenkins & Tripp (2001) used C I excitation to derive the thermal pressure distribution of diffuse Galactic H I clouds, finding that most of the gas resides near $p/k \sim 10^3\text{--}10^4 \text{ cm}^{-3} \text{ K}$, while a smaller fraction occupies a high-pressure tail. Their analysis also showed that unresolved mixtures of gas phases can produce characteristic tracks in the (f_1, f_2) plane, where:

$$f_1 = \frac{N(\text{C I}^*)}{N(\text{C I}) + N(\text{C I}^*) + N(\text{C I}^{**})}, \quad (1)$$

and

$$f_2 = \frac{N(\text{CI}^{**})}{N(\text{CI}) + N(\text{CI}^*) + N(\text{CI}^{**})}. \quad (2)$$

These quantities remain widely used diagnostics of ISM excitation conditions.

To interpret such observations theoretically, Silva & Viegas (2002) developed the **PopRatio** code, a general framework for computing atomic level populations under statistical equilibrium. Their formalism included spontaneous radiative decay, direct radiative excitation, collisional processes, and fluorescence pumping through higher electronic levels. The code was designed to model excitation in both local and cosmological environments, including the effects of the CMB and external UV radiation fields. The statistical equilibrium equations solved by **PopRatio** can be written as:

$$\sum_j n_j Q_{ji} = n_i \sum_j Q_{ij}, \quad (3)$$

where the total transition matrix Q_{ij} contains all radiative and collisional contributions between levels.

Since the development of the original **PopRatio** framework, observational and theoretical studies have increasingly focused on the role of radiation fields in regulating fine-structure excitation. In addition to the Galactic UV field, the extragalactic ultraviolet background (UVB) contributes a diffuse radiation component that becomes particularly relevant in low-density gas and at high redshift. Modern UVB prescriptions, such as those of Khaire & Srianand (2019), provide redshift-dependent estimates of the metagalactic radiation field produced by galaxies and quasars. The UVB intensity evolves strongly with redshift and is expected to play an increasingly important role in the excitation balance of diffuse gas in the CGM and high-redshift absorbers.

At the same time, the CMB itself evolves with redshift according to:

$$T_{\text{CMB}} = 2.728(1+z) \text{ K}, \quad (4)$$

causing direct radiative excitation of fine-structure transitions to become progressively more important at high redshift. This effect has long been recognized as a potential probe of cosmological evolution through fine-structure excitation measurements (Silva & Viegas, 2002). Consequently, disentangling the relative contributions of collisions, local UV radiation, fluorescence, the UVB, and the CMB is essential for accurately interpreting observed level populations.

Motivated by these considerations, we developed **py_popratio**, a Python implementation of fine-structure population calculations inspired by the original **PopRatio** formalism. The code solves the equations of statistical equilibrium for arbitrary multi-level atoms while including collisions, direct radiative excitation, UV fluorescence pumping, Galactic UV fields, and modern extragalactic UV background prescriptions. The modular design enables rapid experimentation with different physical environments, including local ISM conditions, high-pressure clouds, and high-redshift absorbers.

In this work, we describe the physical formalism and numerical implementation of `py_popratio`, validate the code against known limiting cases, and demonstrate its application to CI fine-structure excitation in a variety of astrophysical environments.

2 Methodology of `pypopratio`

`py_popratio` is a Python implementation of atomic fine-structure population calculations inspired by the classical `PopRatio` code of Silva & Viegas (2002). The code computes the relative populations of atomic levels by solving the equations of statistical equilibrium including collisions, direct radiation, and fluorescence (UV pumping). The implementation is modular, with different Python modules responsible for different physical processes.

2.1 1. Atomic Data Input

The module:

`atomic_data.py`

is responsible for reading and parsing atomic data files such as:

`CI.dat`

The primary function:

`parse_atom_file()`

reads:

- Level energies, E_i
- Statistical weights, g_i
- Einstein spontaneous decay coefficients, A_{ij}
- Collision-rate coefficients, $q_{ij}(T)$
- Fluorescence / UV pumping channels

These quantities define the atomic model used throughout the calculation. This follows the philosophy of `PopRatio`, where all atomic quantities are externally supplied and then used to construct transition rates (Silva & Viegas, 2002).

2.2 2. Statistical Equilibrium

The code assumes statistical equilibrium:

$$\frac{dn_i}{dt} = 0, \quad (5)$$

such that all processes populating level i balance all processes depopulating it:

$$\sum_j n_j Q_{ji} = n_i \sum_j Q_{ij}. \quad (6)$$

This is the fundamental equation solved by `py_popratio` and follows the formalism of Silva & Viegas (2002).

The implementation resides in:

`solver.py`

The primary functions are:

`build_Q_matrix()`
`solve_populations()`

2.3 3. Total Transition Rate Matrix

The total transition rate from level $i \rightarrow j$ is written as:

$$Q_{ij} = A_{ij} + B_{ij}u_{ij} + \sum_k n_k q_{ij}^k + \Gamma_{ij}, \quad (7)$$

where:

- A_{ij} is the spontaneous radiative decay rate,
- $B_{ij}u_{ij}$ represents direct radiative excitation or stimulated emission,
- $n_k q_{ij}^k$ are collisional rates with collider k ,
- Γ_{ij} is the fluorescence pumping contribution.

The construction of these terms is performed inside:

`build_Q_matrix()`

within:

`solver.py`

2.4 4. Einstein Coefficients

The Einstein B coefficients are computed from the Einstein A coefficients:

$$B_{ij} = \frac{A_{ij}c^3}{8\pi h\nu_{ij}^3}, \quad (8)$$

with upward absorption rates related through detailed balance:

$$B_{ji} = \frac{g_i}{g_j} B_{ij}. \quad (9)$$

The transition frequency is:

$$\nu_{ij} = c(E_i - E_j), \quad (10)$$

where energies are expressed in cm^{-1} .

These calculations occur inside:

```
build_Q_matrix()
```

2.5 5. Radiation Fields

Radiation fields are implemented in:

```
radiation_fields.py
```

The primary functions are:

```
planck_u()
draine_uv_u_nu()
popratio_uv_u_nu()
ks19_uv_u_nu()
radiation_u_nu()
build_ks19_interpolator()
```

2.5.1 5.1 Cosmic Microwave Background

The CMB is implemented using the Planck blackbody energy density:

$$u_\nu(T_{\text{rad}}) = \frac{8\pi h\nu^3}{c^3} \frac{1}{\exp(h\nu/kT_{\text{rad}}) - 1}, \quad (11)$$

with:

$$T_{\text{rad}} = 2.728(1 + z). \quad (12)$$

This is implemented in:

```
planck_u()
```

and primarily affects direct fine-structure excitation.

2.5.2 5.2 Draine UV Field

The Draine (1978) Galactic UV radiation field is implemented in:

`draine_uv_u_nu()`

The UV field strength is scaled relative to the standard local ISM radiation field:

$$u_\nu = I_{\text{UV}} \times u_{\nu, \text{Draine}}. \quad (13)$$

2.5.3 5.3 PopRatio / Gondhalekar UV Field

The original PopRatio Galactic UV field is implemented in:

`popratio_uv_u_nu()`

following the original PopRatio prescription (Silva & Viegas, 2002).

2.5.4 5.4 Khaire & Srianand UV Background

The extragalactic UV background (UVB) from Khaire & Srianand (2019) is implemented using tabulated spectral intensities:

$$u_\nu(z) = \frac{4\pi}{c} J_\nu(z), \quad (14)$$

where J_ν is the tabulated EBL/UVB intensity.

The functions responsible are:

`build_ks19_interpolator()`

`ks19_uv_u_nu()`

2.6 6. Collisional Excitation and De-excitation

Collisional transition rates are written as:

$$C_{ij}^k = n_k q_{ij}^k(T), \quad (15)$$

where n_k is the density of collider k .

The upward rates are computed using detailed balance:

$$q_{ji} = \frac{g_i}{g_j} \exp \left[-\frac{E_i - E_j}{kT} \right] q_{ij}. \quad (16)$$

These calculations are implemented within:

`build_Q_matrix()`

2.7 7. Fluorescence / UV Pumping

Fluorescence is treated as a two-step process:

$$i \rightarrow u \rightarrow j, \quad (17)$$

where a UV photon excites the atom to an upper electronic state u , followed by radiative decay into another low-lying level.

The UV pumping rate is:

$$K_{iu} = B_{iu} u_\nu(\nu_{iu}). \quad (18)$$

The branching ratio for decay into level j is:

$$P_{u \rightarrow j} = \frac{A_{uj}}{\sum_m A_{um}}. \quad (19)$$

The effective fluorescence rate becomes:

$$\Gamma_{ij} = \sum_u K_{iu} \frac{A_{uj}}{\sum_m A_{um}}. \quad (20)$$

This process is implemented in:

`compute_fluorescence_matrix()`

within:

`solver.py`

2.8 8. Solving the Linear System

The equilibrium equations are rearranged into matrix form:

$$\mathbf{MX} = \mathbf{I}, \quad (21)$$

where the unknowns are the populations relative to the ground level:

$$X_i = \frac{n_{i+1}}{n_1}. \quad (22)$$

The linear system is solved using:

`solve_populations()`

After solution, the populations are normalized such that:

$$\sum_i n_i = 1. \quad (23)$$

2.9 9. Observable Population Ratios

The code computes observable ratios such as:

$$\frac{N(\text{CI}^*)}{N(\text{CI})} \simeq \frac{n_1}{n_0}, \quad (24)$$

and:

$$\frac{N(\text{CI}^{**})}{N(\text{CI})} \simeq \frac{n_2}{n_0}. \quad (25)$$

The Jenkins & Tripp (2001) fractional populations are also commonly used:

$$f_1 = \frac{N(\text{CI}^*)}{N(\text{CI}) + N(\text{CI}^*) + N(\text{CI}^{**})}, \quad (26)$$

$$f_2 = \frac{N(\text{CI}^{**})}{N(\text{CI}) + N(\text{CI}^*) + N(\text{CI}^{**})}. \quad (27)$$

Population-ratio grids are generated through:

`compute_population_ratios()`

2.10 10. Physical Interpretation

The resulting level populations can be used to constrain:

- Hydrogen density, n_{H}
- UV radiation field intensity, I_{UV}
- Thermal pressure:

$$\frac{p}{k} \simeq n_{\text{H}} T. \quad (28)$$

The code therefore enables direct comparison between observed absorption-line column densities and theoretical excitation models for the ISM, CGM, and high-redshift absorbers.

References

- Draine, B. T. 1978, ApJS, 36, 595
- Jenkins, E. B., & Tripp, T. M. 2001, ApJS, 137, 297
- Khaire, V., & Srianand, R. 2019, MNRAS
- Silva, A. I., & Viegas, S. M. 2002, MNRAS